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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-060

1. Topic (12 words max)

AI Based Smart Retail Recommendation System

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references (300 words max) – references not included in word count.

With the retail industry undergoing rapid transformation aligned with digitalization, e-commerce and mobile shopping have experienced significant growth, driven by smart retail technologies and artificial intelligence (AI) adoption. Leading digital platforms such as Amazon and Daraz utilize advanced AI-based systems for personalized recommendations, demand forecasting, and customer engagement, delivering highly tailored shopping experiences that enhance conversion rates and operational efficiency [1]). In contrast, physical retail stores—particularly in developing economies like Sri Lanka—continue to rely on manual analysis methods such as customer walk-in counts and basic sales reports, lacking real-time customer behavior insight and data-driven decision-making processes [2]

This technology gap limits retail customers' ability to discover the best offers, miss relevant products, and ultimately purchase fewer intended items because they do not receive personalized guidance or recommendations that match their preferences [3]. Moreover, store owners and managers are obligated to monitor stores manually to understand shopper behavior, which increases operational workload and leads to inefficient product placement and inventory planning. Without analytics systems, retailers also struggle to identify irrelevant or slow-moving items on shelves, reducing profitability and increasing waste. Furthermore, retail chains and product manufacturers lack detailed insights into product performance in physical environments, hindering demand forecasting and supply chain optimization across outlets.

The consequences of this digital lag are not only economic but also social, ethical, environmental, and legal. Customers—especially the elderly and disabled—struggle without appropriate assistance or guidance in stores, contributing to disparities in the shopping experience. Retailers in these markets risk losing customers due to poor service quality and the absence of personalized recommendations,

affecting revenue and competitiveness. From an ethical standpoint, as AI systems are introduced to retail, issues such as consumer privacy, fairness, and data transparency become increasingly important in ensuring trust and regulatory compliance [4] Additionally, inefficient inventory practices cause overstocking and wastage of goods, leading to environmental impacts due to unused products and non-renewable waste [2].

The need for machine learning, computer vision, big data analytics, and recommendation algorithms to support IoT-based smart services and AI-enabled business solutions in retail is thus critical to closing this gap. These technologies can improve economic outcomes, support innovation and infrastructure development, and promote responsible consumption and production—advancing several United Nations Sustainable Development Goals, including Decent Work and Economic Growth (8), Industry, Innovation, and Infrastructure (9), and Responsible Consumption and Production (12) by enabling efficient stock management and reducing waste.

References :-

[1] O. K. Oladele, AI in Retail: Personalized Shopping Experiences, Inventory Management, and Supply Chain Optimization, Mar. 2025.

[2] AI in Retail | IBM, IBM, 2025.

[3] D. Patil, Artificial intelligence in retail and e-commerce: Enhancing customer experience through personalization, predictive analytics and real-time engagement, Nov. 2024.

[4] A. Adanyin, Ethical AI in Retail: Consumer Privacy and Fairness, European Journal of Computer Science and Information Technology, Jul. 2024

6. Brief description of Existing Research and Systems (300 words max)

All most all retail recommendation systems are designed for online shopping platforms rather than physical stores. Online platforms , use browsing history , purchase history and user profiles to generate personalized product recommendations. In physical retail stores , shop owners do not process user profiles of customers , but some advanced stores use heatmaps from cameras to see foot traffic however these footage cannot be integrated to the system. Unlike in an online store , retail stores do not have a technology where stocks are updated

Computer vision was used to track customer movements and generate heatmaps to understand which shelves attract attention[1]

Machine Learning based recommendation engines improve sales and customer satisfaction in e commerce [2]

Another project was launched where smart shelf cameras were installed along with deep learning to detect , product picking , empty shelves and customer engagement

Gaps Identified :-

No integration :- A real time intelligent system is needed

Focus on online retail :- Physical retail environments are largely ignored , hence less importance is given to physical retail systems and more focus on online platforms

Lack of Affordable solutions :- Some retail stores are unable to install one camera per shelve due to high costs.

References :-

[1] Zhang, Y., Wang, S., & Chen, X. (2022). *Computer vision–based customer behavior analysis in smart retail*. **IEEE Access**, 10, 112345–112360. https://doi.org/10.1109/ACCESS.2022.3145678

[2] Li, X., & Karahanna, E. (2021). *AI-based recommendation systems in retailing: A review and research agenda*. **Decision Support Systems**, 142, 113471.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed AI-Based Smart Retail Recommendation System is designed to bridge the gap between online recommendation systems and physical retail environments. Unlike traditional retail analytics that rely only on sales data or isolated camera heatmaps, this solution introduces a real-time, integrated, and intelligent retail platform that combines customer behavior, emotions, operations, checkout data, and inventory insights into a single system.
This research directly addresses the gaps identified in Section 6 by shifting the focus from online-only recommendation systems to physical retail stores, where customer profiles are usually unavailable. Instead of relying on browsing history or login-based user data, the proposed system uses computer vision, artificial intelligence, and machine learning to infer customer intent, preferences, and engagement in real time using non-intrusive in-store data sources.
The system integrates CCTV-based customer behavior analysis, emotion detection, smart inventory tracking, checkout automation, and personalized recommendation logic. By doing so, it overcomes the lack of integration seen in existing solutions and enables store managers and higher management to access unified insights through a single platform. This integration allows informed decision-making related to marketing strategies, store layout optimization, staffing, and inventory replenishment.
To address affordability concerns, the proposed system avoids the need for installing a camera for every shelf. Instead, it leverages existing CCTV infrastructure, edge-based processing, and selective vision analysis to reduce hardware and operational costs, making the solution suitable even for small and medium-sized retail stores.

How the Proposed System Addresses Identified Gaps

- Integration Gap:

Combines customer behavior, emotion analytics, checkout performance, and inventory data into one unified system.

- Online-Centric Focus Gap:

Enables recommendation and analytics in physical retail stores without requiring user login or profiles.

- Cost and Scalability Gap:

Uses existing cameras and AI-based inference instead of expensive per-shelf hardware installations.

System Perspective: Input – Process – Output

Inputs

- Live CCTV video feeds
- Point-of-sale (POS) and checkout transaction data
- Inventory and stock-level data
- Environmental data (store layout, time, foot traffic)

Processing

- Computer vision for customer detection, movement tracking, and emotion analysis

- Machine learning models

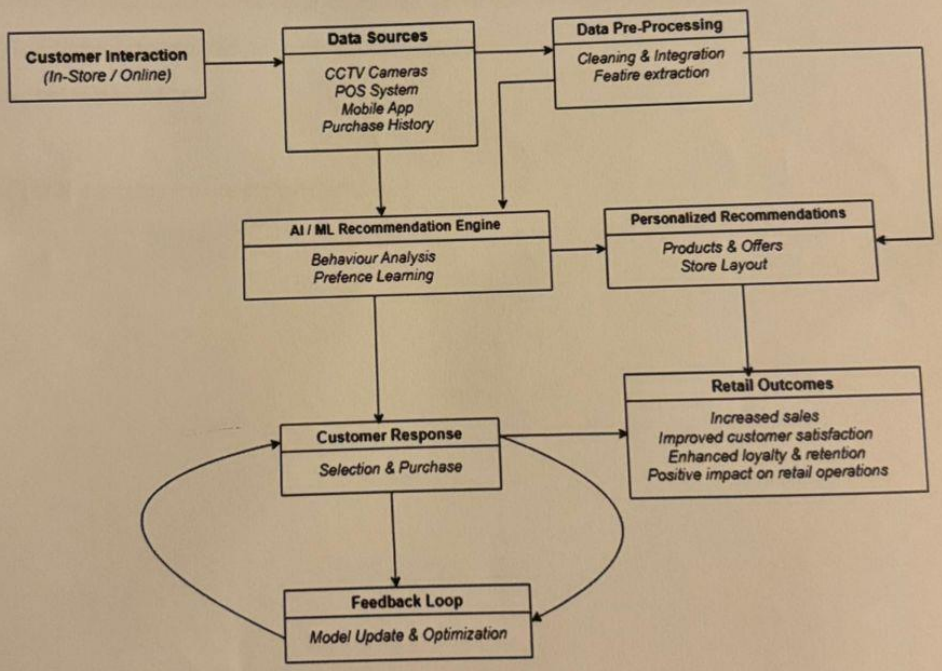
for behavior analysis and recommendation generation

- Real-time data fusion across operations, checkout, and inventory modules
- Decision logic for personalization and operational optimization

Outputs

- Personalized product and promotion recommendations
- Real-time alerts for customer dissatisfaction or operational bottlenecks
- Inventory replenishment and shelf optimization insights
- Management dashboards with actionable business intelligence

Conceptual Diagram



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

- *How can members gain essential specialized domain expertise?*

By visiting retail stores to observe the customer flow and shelf behavior, conduct short interviews, review as studies and existing research material, studying about customer theories

- *How can a member gain the technical knowledge and skills required for the project?*

Machine Learning, Open-source libraries, Python, proto type development and testing

- *How is the data required for the project collected?*

From Cool Planet Pvt Ltd

- *Does this project require ethical clearance?*

9. Objectives and Novelty

Main Objective

Emotion Aware Shelf Level Real Time Recommendation Engine for Physical Retail
To capture and analyze in store customer behavior
Link emotional responses at shelves
Build real time AI recommendation engine
Improve in store decision making for retailers
Evaluate the effectiveness of emotion aware recommendations

Member Name with Registration No	Sub Objective	Tasks	Novelty
WIJETUNGA J A R K IT22146588	***Personalized Marketing Intelligence*** • *Customer–Item–Promotion Interaction Analytics* • *Customer Feedback and Hidden Demand* • *Promotion Attribute Engine* • *Real Time Promotion Performance Dashboard* • *Promotion Optimization* • *Emotion-Aware Visual Adjustment*	• ***Customer–Item–Promotion Interaction Analytics*** *Builds a customer item promotion interaction graph. (Answers the question : Did the customer buy because of the discount ?) . This feature tracks how the customers react to the products , discounts and promotions.* • ***Customer Feedback and Hidden Demand*** *(Example:- A customer wants to buy an item of her/his choice with the intended material and color. However, the customer is unaware of the availability. The customer has the facility to fill in and upload a form. The request will appear in the internal dashboard. Upon checking the availability from the inventory department staff will reply to the query. Upon the*	• ***Customer–Item–Promotion Interaction Analytics*** *Existing systems in the retail industry the tracking process tracks only the purchase. However , with the proposed system the management will be able to track the customer categories , promotion purchases and the effectiveness* • ***Customer Feedback and Hidden Demand*** *Existing platforms do not track the customer's demand. But get the customer request as a query and discarded when responded. The proposed system will pin the request (if the customer's requested item is not in stock) as a hidden demand which will give an idea to the product owners an idea about the products which have higher demands*

		customer's request the item could be delivered or made available for pick up. However if the requested item is not in stock the customer will be notifies with regards and the customer's request will be added to a hidden demand section) • ***Promotion Attribute Engine*** *Since a retail business could have many promotions and discounts for a wide range of chosen items . It important for the management to identify which promotions caused higher sales and which promotions did not work out* • ***Real Time Promotion Performance Dashboard*** *The proposed system will show live , the offers which are working , ignored shelves , failing promotions and customers response.* • ***Promotion Optimization*** *Evaluates promotion success and adjust future discounts and bundles (What will increase profit ant not just sales) This feature will help the management to focus on the shelves which are ignored and promote more discounts* • ***Emotion-Aware Visual Adjustment***	• ***Promotion Attribute Engine*** *Existing systems capture the number of sales due to promotions and not in detail. The proposed system will be able to capture the profit by specific promotion as there will be credit scores for different promotions .* • ***Real Time Promotion Performance Dashboard*** *Existing retail dashboards show the sales number while the proposed system shows the behavioral effectiveness of promotions. The system integrate the inventory status of the items as well* • ***Promotion Optimization*** *Existing systems will only focus on rule base while the proposed system will give predictions with regards to promotions and discounts* • ***Emotion-Aware Visual Adjustment*** *Existing systems do not have a feature for generating banners, except for online shopping platforms . The proposed system will be able to generate banners according to the targeted customer category*

		Designing promotional banners. Since the customer behavior is monitored in the proposed system, the banners will be generated accordingly (Example :- If the promotion is for ladies , the banner will associate more feminine colors in the banner)	
PERERA L A D A S IT22243812	***Customer Emotions and Behavior Detection***	• ***Emotion analytics and behavioral pattern analysis.*** *For example Cameras in a retail store notice that some customers look confused or frustrated while browsing. The system identifies these emotional patterns and helps staff understand where customers usually face difficulties.* • ***Real-time monitoring and evaluation of customer interactions*** • ***Predictive Customer Intent & Experience Risk Modeling.*** *For Example: The system predicts that a customer may leave the store without buying because they appear annoyed and have been waiting too long. Staff are alerted early to step in and help.* • ***Context-Aware Behavioral Anomaly Detection.*** *For Example: If a customer keeps walking back and forth in one area for a long time, the system notices this unusual behavior and checks whether the customer needs help or if there is a problem.*	• *AI-driven fusion of visual and behavioral cues for high-precision emotional intelligence.* *Emotions patterns will be categorized into four emotions namely happy , sad , angry and confused and behavioral patterns will be categorized as calm , excited , frustrated* • *Temporal emotion tracking for proactive customer experience management.* • *Emotion-driven predictive intelligence for preventive service optimization.* • *Context-adaptive behavior anomaly detection using spatio-temporal analytics.* • *Emotion-responsive personalization powered by real-time affective computing.*

		• ***Emotion-Driven Personalized Engagement Recommendation Engine.*** *For Example: When the system detects that a customer looks confused, it suggests that a staff member approach the customer and offer guidance or product information.*	
SILVA T A IT22284952	***Retail Operations and Checkout Automation***	• ***Real-Time Checkout Process Analytics & Detection .*** *EX: For instance During busy evening hours in a large supermarket, the system notices that two checkout counters are working slower than others because barcodes are not scanning properly. The system immediately highlights these counters and alerts the floor manager, who redirects customers to faster lanes or fixes the issue.* • ***Integrated In-Store Behavior and Checkout Flow Correlation.*** *EX: For instance CCTV and footfall sensors show customers clustering near promotional aisles but abandoning baskets when checkout queues exceed five people. The system correlates dwell time in aisles with queue length data and identifies checkout congestion as the main cause of lost sales.* • ***Automated Exception Handling and Process Automation.*** *EX: At a self-checkout kiosk, a customer scans an item with a price mismatch. The system*	• *Real-time checkout intelligence using live transaction and queue dynamics.* • *A unified behavioral–transactional analytics framework for holistic retail operations optimization.* • *Autonomous checkout exception management with minimal human intervention.* • *Next-generation frictionless checkout powered by behavioral intelligence.* • *Single-pane-of-glass visibility for intelligent retail operations management.*

		automatically verifies the correct price from the inventory database and resolves the issue without staff involvement, reducing transaction delay. • ***Intelligent Self-Checkout and Frictionless Payment Enablement.*** *EX: In a cashier-less convenience store, the system detects normal customer behavior during item selection and payment, enabling seamless walk-out checkout. If abnormal behavior is detected, the system discreetly switches to verification mode without interrupting other customers.* • ***End-to-End Retail Operations Intelligence Dashboard***	
METHSARA W M IT22244598	***Smart Inventory Optimization*** • *To intelligently manage and optimize inventory levels across all branches and sales channels* • *To predict future product demand using historical sales data and in-store customer behavior patterns* • *To minimize stockouts, overstock, and slow-moving inventory in the omni-retail environment* • *To automate*	• **Collect and integrate multi-source inventory and sales data** This task involves gathering data from POS systems, branch stock records, online sales platforms, and warehouse inventory. Data is cleaned and combined into a unified dataset. Example: consolidating sales of ladies' handbags from Colombo, Kandy, and Online Store into one dashboard. • **Analyze historical sales trends and seasonal demand patterns** Past sales behavior is studied to understand peak seasons, festival trends, and slow periods. This analysis helps identify items that	• ***Integration of CCTV-based customer behavior analytics*** *Uses face recognition and shelf-interaction data to understand real customer interest, not just sales. Helps estimate demand before purchase happens.* • ***Dynamic and adaptive inventory optimization*** *Reorder levels and safety stock automatically adjust with seasons, promotions, and footfall. Inventory decisions continuously learn and update in real time.* • ***Intelligent cross-branch stock reallocation*** *Detects excess stock in one branch and shortages in another. Recommends internal transfers before new purchasing.*

	replenishment decisions, reorder quantities, and safety stock levels • *To support managerial decision-making using data analytics and real-time inventory insights*	behave differently during specific months. Example: identifying that winter jackets sell more during November–January, and school bags peak before school opening periods. • **Perform automated ABC/XYZ inventory classification** Products are categorized based on sales frequency, value, and demand variability. This helps prioritize attention to high-value, fast-moving items and control slow movers. Example: Classifying T-shirts as A-high priority, belts as B-medium, and scarves as C-low priority items. • **Forecast product demand at branch level** Predictive models are used to estimate future demand for each product in each branch. This forecasting considers sales history, trends, and customer behavior insights. Example: predicting higher demand for sportswear in branches located near gyms or universities. • **Determine dynamic safety stock, reorder point, and optimal order quantity** The system automatically calculates when to reorder and how much to reorder, based on demand variation and lead time uncertainty.	• ***Early detection of slow-moving and high-interest items*** *Compares product views and handling with actual sales to find items customers like but don't buy. Enables early actions like price or display changes.* • ***Unified omni-retail inventory management*** *Optimizes stock across online and physical stores together in one system. Prevents channel-wise stock mismatches.* • ***Context-aware design for Cool Planet and Sri Lankan market*** *Considers local seasons, festivals, climate, and regional demand patterns. Makes the solution practically suitable for real Sri Lankan retail operations.*

		Example: automatically generating a reorder alert when ladies' jeans reach safety stock levels and recommending exact reorder quantity. • **Identify overstock, stockout risk, and recommend stock reallocation** Stock levels are continuously monitored to detect excess inventory, low stock, and non-moving products. The system suggests branch-to-branch transfers before new purchasing. Example: shifting extra shoes from a low-demand branch to a high-demand branch instead of buying new stock.	

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10. **This part is to be filled by the Supervisor and the Co-supervisor of the Project.**

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes		No	✓

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	✓

e) Any other comments:

Accept with minor corrections -

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Dilanka	Wijendra	2026/01/12
Co-Supervisor	Ms.	Jenny	Krishara	Jenny 2026/01/12
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				